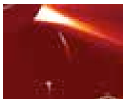
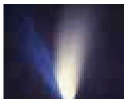


**The Sungrazer Project**
Join amateur astronomers to hunt for new comets by analyzing telescope images from the ESA/NASA mission. **http://dgit.in/mar21-11**



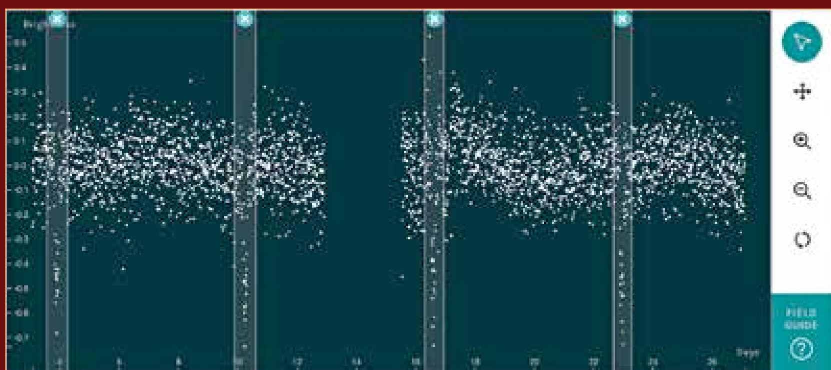
**Modern Astronomical Discoveries**
Central Bureau for Astronomical Telegrams, announces new astronomical discoveries.
**http://dgit.in/mar21-12**

planet Hunters have found over 100 candidate exoplanets that the computers and official image processing pipelines missed. The project still continues, with citizen scientists now analysing data from the Transiting Exoplanet Survey Satellite (TESS) telescope, that has a field of view 400 times larger than Kepler, and is going to survey 85 percent of the sky. The scientists themselves have not manually gone through the Kepler data, like the citizen scientists have. The team from Yale publishes the findings of new discoveries from the Planet Hunters projects, and include the citizen scientists as co-authors in the papers.

Astronomers from the Las Cumbres Observatory have also setup a site for citizen scientists and even school children to analyse the data captured by their telescopes in Hawaii, Australia and California. Interested participants have to choose stars that they would like to monitor, calibrate the star of interest with other stars of similar size and brightness, then look at the light curves for the telltale dips of exoplanet transits. The telescopes conduct follow up observations of potential exoplanet candidates flagged by other instruments, and the website allows you to easily access and view the transit events and light curves for some of these candidates.

## SOHO COMET HUNTERS

The Solar and Heliospheric Observatory (SOHO) was deployed in 1995 with the objective of observing the Sun, study the coronal activity, and allow scientists to predict space weather. The real time data from the instrument is available to the general public, with various third party sites providing additional processing and access. This means that anyone, anywhere in the world can look at what the spacecraft is seeing, and one of the pleasant bonuses of the satellite is that it is very, very good at spotting comets. Over the course of its 25 year mission, over 4,000 new comets have been identified, most by comet hunters who just look at the live data as it is streaming in. The raw information is released in realtime, and only the most experienced comet hunters work with the raw file format which is an obscure one called FITS. There is a community of comet hunters that NASA supports through an initiative known as the Sungrazer project. Once a comet is flagged by citizen scientists, researchers report the sighting to the Central Bureau for Astronomical Telegrams, a body that makes announcements for the International Astronomical Union, which then goes on to officially name the object.



Credit: PHAD

**The Planet Hunters interface, with several potential exoplanet transit events marked. The spots mark levels of brightness in the stars, the larger dips, or lower dots indicate potential planets that are likely to be larger.**

## ZOONIVERSE

Zooniverse has now become a citizen science portal and the host for several projects, including Exoplanet Hunters. Zooniverse grew out of Galaxy Zoo, a project that still continues. The Sloan Digital Sky Survey had captured images of over 900,000 galaxies, and scientists wanted to understand more about this population. They hired volunteer scientists, similar to teachers who volunteer during the census, and together managed to classify all the galaxies within 75 days. The researchers had expected between 20,000 to 30,000 volunteers, but over a 100,000 got to work, accelerating the process. It took the Galaxy Zoo project months to conduct the largest galaxy census, instead of the years it would take for scientists to classify all the galaxies. The project continues to this day, and users are only given an image and a series of questions to answer. The volunteers have to look at the presented images, and classify the galaxies as barred, spiral, irregular or spherical. There are other interesting astronomy projects now hosted on Zooniverse, which grew out of Galaxy Zoo. The SpaceWarps project invites citizen scientists to identify instances of gravitational lensing. Over 35,000 volunteers have already identified 30 lenses that had been missed by the computers. Another project that will keep you busy forever is the Milky Way Project, where users have to look at pictures of pretty nebule, identify stellar wind bubbles as well as the star from which they originate. Users have to draw the bubbles on the interface, or portions of the bows, and identify the stars where the gas and radiation originated from. No matter what aspect of the skies interest you, there is probably a citizen science astronomy project that you can participate in. d